BEML/RS (3R)/T&C/RAMS/1246
Project Ref: 1249-21-1246

Date: 18.08.2021

The Project Manager (RS),
General Consultant, MYCEL,
KMRCL Bhavan (HRBC Office Complex),
Munshi Premchand Sarani,
Kolkata- 700021

Kind Attn: Shri. Navendu Lad

Dear Sir,

Sub: Contract RS (3R): Observations towards methodology for calculation of RAMS data

Ref.: GC/KMRC letter1149-21-746 dated: 16.08.2021

Following are the observations made as regards to methodology used while calculating RAMS data in the report which has been forwarded vide the letter referred above.

A. Document Approval:

Unapproved document no GR/TD/3457 Rev – 0 has been referred for the reliability calculation whereas the approved document is rev-1.

B. MDBF Calculation:

Mean Distance Between Failures is to be calculated by dividing (X) by (Y) where

X is the sum of the KMs run by the revenue trains under consideration and

Y is the number of service failures encountered by the same set of trains during the period 7th month to 24th month of the revenue service.

The data accompanied by the referred letter consists of a number of train withdrawals with insufficient reasons. Case wise justification for countering the withdrawals has been attached in annexure-I and the same is being submitted as a part of the weekly progress report every week.

The data accumulated by BEML suggests MDBF achieved till date is above 1,43,725 KMs.

C. MDBCF Calculation:

1. Defined Mean Distance between (2 consecutive) failures of any component in a 3-Car unit (Say X KM) indicates that there is acceptable probability of 1 failure among 3 cars every defined distance (X KM). Thereby, the acceptable probability of failures becomes 2 for every 6 cars within the defined distance.

Hence proportionate MDBCF (KMs) for 6 car set is to be divided by 2 whereas it has been multiplied by 2 in the forwarded report.

2. Further, as the name suggests, MDBCF is calculated for the distance covered by any **component before failure**. In the report forwarded with respective datasheet confirms that even air leakage without failure of any component has been counted as COMPONENT FAILURE.

3. It is also understood that KM attained by complete 6 car train is considered as total distance covered by 14 VACs equipped in the train instead of multiplying the Train KMs with population of the similar components equipped in the train.

D. MTTR:

MTTR is required to be calculated over a period of time with basic consideration that the trains may clock 60,000 KM during 2nd six months' period and 1,00,000 KM in 2nd year of service. Actual KM clocked by the trains is too low as compared to the basic consideration. Hence, it is too early to declare MTTR at this stage.

In the view of the above, kindly refer to the MDBF data and MDBCF data as indicated in the "Train Reliability indicator" sheet being submitted as parts of weekly status which are being submitted as an indicative information of the ongoing process.

Thanking You

Yours Sincerely
(For BEML Limited)

(P. Poongumaran)
Project Manager
KMRCL RS(3) Project

Cc: 1. Shri Ken Wong, Project Director, GC/MYCEL, KMRCL
2. Shri Prosenjit Chakraborty Chief Engineer (Electrical)-I, KMRCL